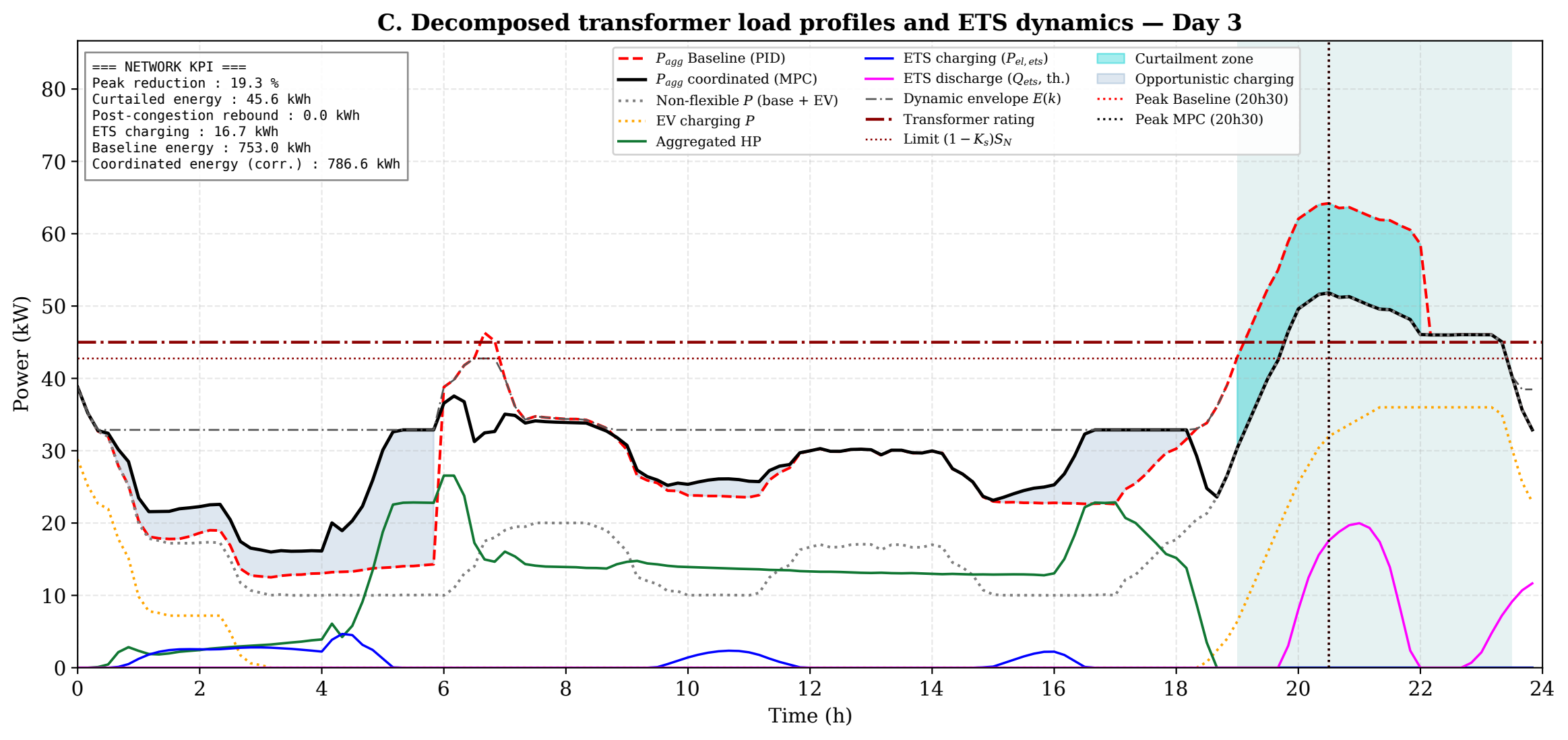
C. Decomposed transformer load profiles and ETS dynamics — Day 3
=== NETWORK KPI ===
Peak reduction : 19.3 %
Curtailed energy : 45.6 kWh
Post-congestion rebound : 0.0 kWh
ETS charging : 16.7 kWh
Baseline energy : 753.0 kWh
Coordinated energy (corr.) : 786.6 kWh
P_{agg} Baseline (PID)
P_{agg} coordinated (MPC)
Non-flexible P (base + EV)
EV charging P
Aggregated HP
ETS charging ($P_{el,ets}$)
ETS discharge (Q_{ets}, th.)
Dynamic envelope $E(k)$
Transformer rating
Limit $(1-K_s)S_N$
Curtailment zone
Opportunistic charging
Peak Baseline (20h30)
Peak MPC (20h30)
Power (kW)
Time (h)
0
10
20
30
40
50
60
70
80
0
2
4
6
8
10
12
14
16
18
20
22
24